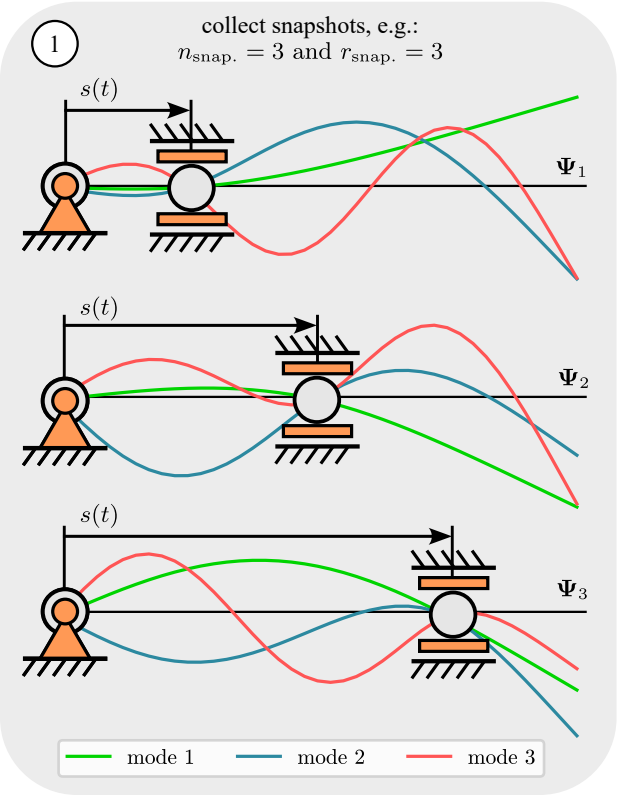


2 assemble *pool*

$$\Psi_{\text{all}} = [\Psi_1, \Psi_2, \Psi_3]$$

$$\Psi_{\text{all}} \in \mathbb{R}^{\text{nDOF} \times (n_{\text{snap.}} r_{\text{snap.}})}$$

3 perform SVD

$$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^* = \text{SVD}\left(\Psi_{\text{all}}\right)$$

4 select glob. modes

$$\Psi_{\text{glob.}} = \boldsymbol{U}_{:,1:r_{\text{glob.}}}$$

$$\Psi_{\text{glob.}} \in \mathbb{R}^{\text{nDOF} \times r_{\text{glob.}}}$$

5 reduce system (const. matrices)

$$\boldsymbol{M}_{\text{red.}} = \Psi_{\text{glob.}}^{\text{T}} \boldsymbol{M} \Psi_{\text{glob.}}$$

$$\boldsymbol{K}_{\text{red.}} = \Psi_{\text{glob.}}^{\text{T}} \boldsymbol{K} \Psi_{\text{glob.}}$$

6 set up Lagrangians Type 1

$$\boldsymbol{f}_{\text{red.}}(t) = \Psi_{\text{glob.}}^{\text{T}} \boldsymbol{f}(t)$$

$$\boldsymbol{C}_{q,\text{red}}\left(s(t)\right) = \boldsymbol{C}_q\left(s(t)\right) \Psi_{\text{glob}}$$

$$\boldsymbol{g}\left(s(t), \boldsymbol{q}(t)\right) = \boldsymbol{0}$$

$$\boldsymbol{M}_{\text{red.}} \ddot{\boldsymbol{p}}(t) + \boldsymbol{K}_{\text{red.}} \boldsymbol{p}(t) + \boldsymbol{C}_{\text{q,red.}}^{\text{T}}\left(s(t)\right) \boldsymbol{\lambda}(t) = \boldsymbol{f}_{\text{red.}}(t)$$